

PRODUCTION PLANNING AND CONTROL

WHAT IS PPC?

*“The highest efficiency in production is obtained by manufacturing the required quality of product , of required quantity ,at the required time by the best and cheapest method” - Hence, **PPC is a tool to coordinate all manufacturing activities in a production system.***

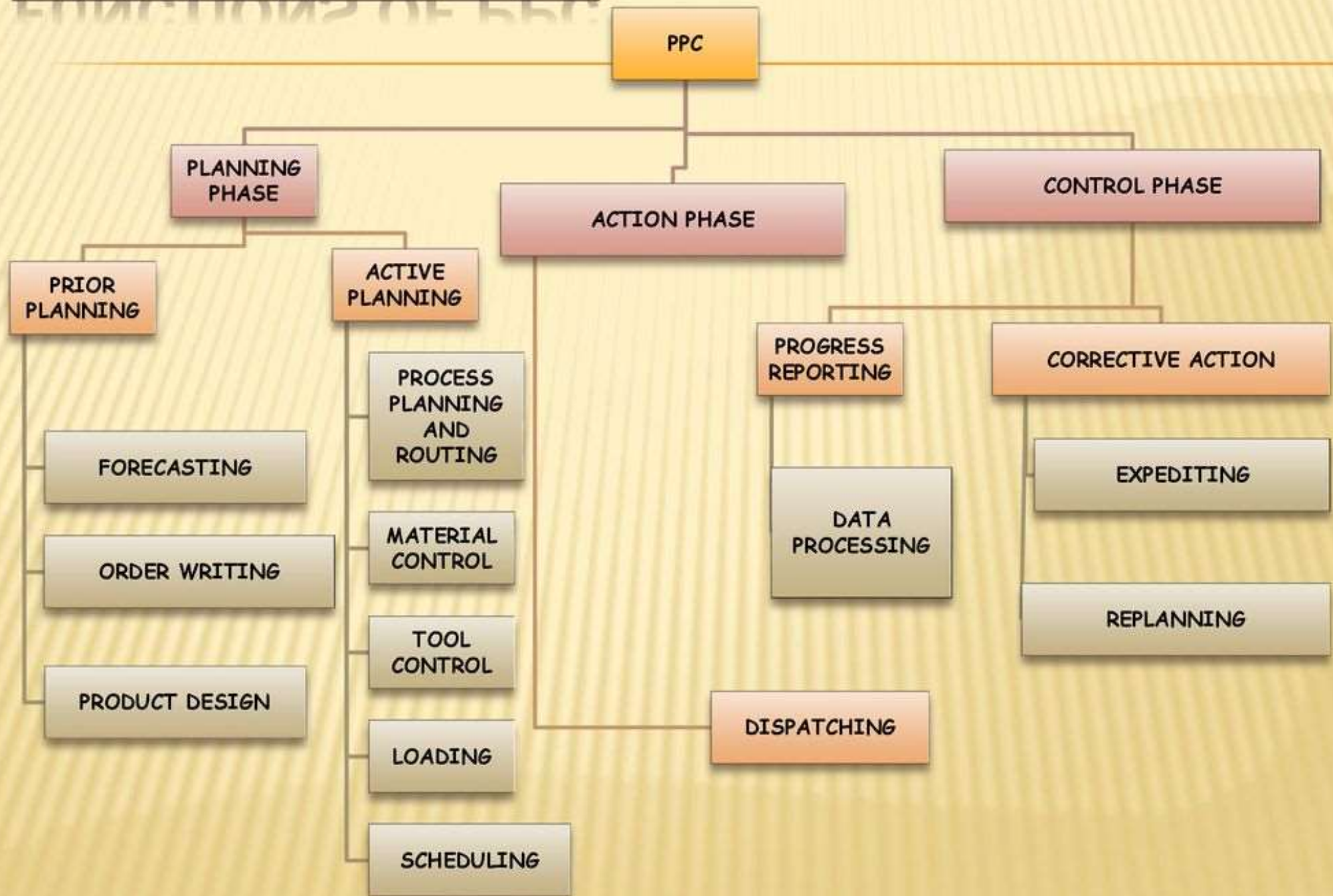
OBJECTIVES OF PPC

- ✘ To deliver required goods in required quantities to the customer in the required delivery schedule to achieve maximum customer satisfaction and minimum cost
- ✘ To ensure maximum utilization of the available resources
- ✘ To ensure production of quality products
- ✘ To minimize the manufacturing time

CONTINUED.....

- ✘ To maintain optimum inventory levels
- ✘ To maintain flexibility in manufacturing operation
- ✘ To coordinate between labor and machines and various supporting documents
- ✘ To plan for plant capacities for future requirements
- ✘ To remove bottle neck at all stages of production and to solve problems related to production
- ✘ To ensure effective cost reduction and cost control

FUNCTIONS OF PPC



PLANNING PHASE

Estimation of type, quality & quantity of future work

Giving authority to one or more person to undertake a particular job

Collection of information regarding specifications, BOM, drawings etc

Finding the most economical process of doing a work and deciding where and how work will be done

Involves determining the requirements and control of materials

Involves determining the requirements and control of tools used

Assignment of work to manpower, machinery etc

Determines when and in what sequence the work will be carried out. It fixes the starting as well as ending time for the job

PRIOR PLANNING

FORECASTING

ORDER
WRITING

PRODUCT
DESIGN

ACTIVE PLANNING

PROCESS
PLANNING
AND ROUTING

MATERIAL
CONTROL

TOOL
CONTROL

LOADING

SCHEDULING

ACTION PHASE

It is the transition from planning to action phase. In this phase the worker is ordered to start the work

ACTION PHASE

DISPATCHING

CONTROL PHASE

Data regarding the job process is collected

It is interpreted with the present level of performance

Taking action if the progress reporting indicates the deviation of the plan from the originally set targets

Re planning of the whole affair becomes essential, in case expediting fails to bring the deviated plan to its actual path

PROGRESS
REPORTING

DATA
PROCESSING

CORRECTIVE
ACTION

EXPEDITING

REPLANNING

STAGES/STEPS IN PPC



ROUTING

- Related to production planning



SEQUENCING

- Related to production planning



SCHEDULING

- Related to production control



DISPATCHING

- Related to production control



FOLLOW UP

- Related to production control

1. ROUTING

- ✘ Routing is the first step in production planning and control.
- ✘ Routing can be defined as the process of deciding the path (route) of work and the sequence of operations.
- ✘ In short, routing determines 'What', 'How much', 'With which', 'How' and 'Where' to produce.

ADVANTAGES OF ROUTING

- ✖ Routing gives a very systematic method of converting raw-materials into finished goods.
- ✖ It leads to smooth and efficient work.
- ✖ It leads to optimum utilization of resources; namely, men, machines, materials, etc.
- ✖ It leads to division of labor.
- ✖ It ensures a continuous flow of materials without any backtracking.

CONTINUED.....

- ✗ It saves time and space.
- ✗ It makes the work easy for the production engineers and foremen.
- ✗ It has a great influence on design of factory's building and installed machines.

STEPS /PROCEDURE OF ROUTING

Product analysis determines what to manufacture and purchase

Product analysis is done again to determine materials required for production

Fix the manufacturing operations and their sequences

Decide the number of units to be manufactured in each lot of production

Estimate the margin of scrap in each lot of production

Analyse the production cost

Prepare the production control forms for effective routing

Prepare a separate route sheet for each order

TECHNIQUES OF ROUTING

- ✕ Route card
- ✕ Work sheet
- ✕ Route sheet

1. ROUTE CARD

- ✘ This card always accompanies with the job throughout all operations.
- ✘ This indicates the material used during manufacturing and their progress from one operation to another.
- ✘ In addition to this the details of scrap and good work produced are also recorded.

2. WORKSHEET

It contains

- Specifications to be followed while manufacturing.
- Instructions regarding routing of every part with identification number of machines.
- This sheet is made for manufacturing as well as for maintenance.

3. ROUTE SHEET

- It is also called as route card
- It lists the manufacturing operations in the decided sequence along with the machines associated with each operation
- It also indicates the department in which the operation is to be done and the part will go for the next operation
- It also consists of the information such as part name, part number and product number
- It gives information about the material specification and cutting tools, jigs, fixtures and necessary devices for each operation.

2. SEQUENCING

- ✖ Defined as the order in which jobs pass through machines or work stations for processing
- ✖ The main aim is to find out such sequence out of the possible sequence that will complete the work in shortest time
- ✖ Sequencing problems becomes tedious as the number of jobs and machines increases

3. SCHEDULING

Scheduling means setting of starting and finishing dates for each operation, assembly and the finished product.

It also means to :

- ✗ Fix the amount of work to do.
- ✗ Arrange the different manufacturing operations in order of priority.
- ✗ Fix the starting and completing, date and time, for each operation

4. DISPATCHING

- ✖ It's the next step after scheduling
- ✖ Also means starting the actual production of a particular work which has been planned in routing schedule.
- ✖ It provides the necessary authority to start the work.
- ✖ It is based on route-sheets and schedule sheets.

Dispatching includes the following:

- ✘ Issue of materials, tools, fixtures, etc., which are necessary for actual production.
- ✘ Issue of orders, instructions, drawings, etc. for starting the work.
- ✘ Maintaining proper records of the starting and completing each job on time.
- ✘ Moving the work from one process to another as per the schedule.
- ✘ Starting the control procedure.
- ✘ Recording the idle time of machines.

4. FOLLOW UP

- ✘ Follow-up or Expediting is the last step in production planning and control. It is a controlling device. It is concerned with evaluation of the results.
- ✘ Follow-up finds out and removes the defects, delays, limitations, bottlenecks, loopholes, etc. in the production process. It measures the actual performance and compares it to the expected performance. It maintains proper records of work, delays and bottlenecks. Such records are used in future to control production.

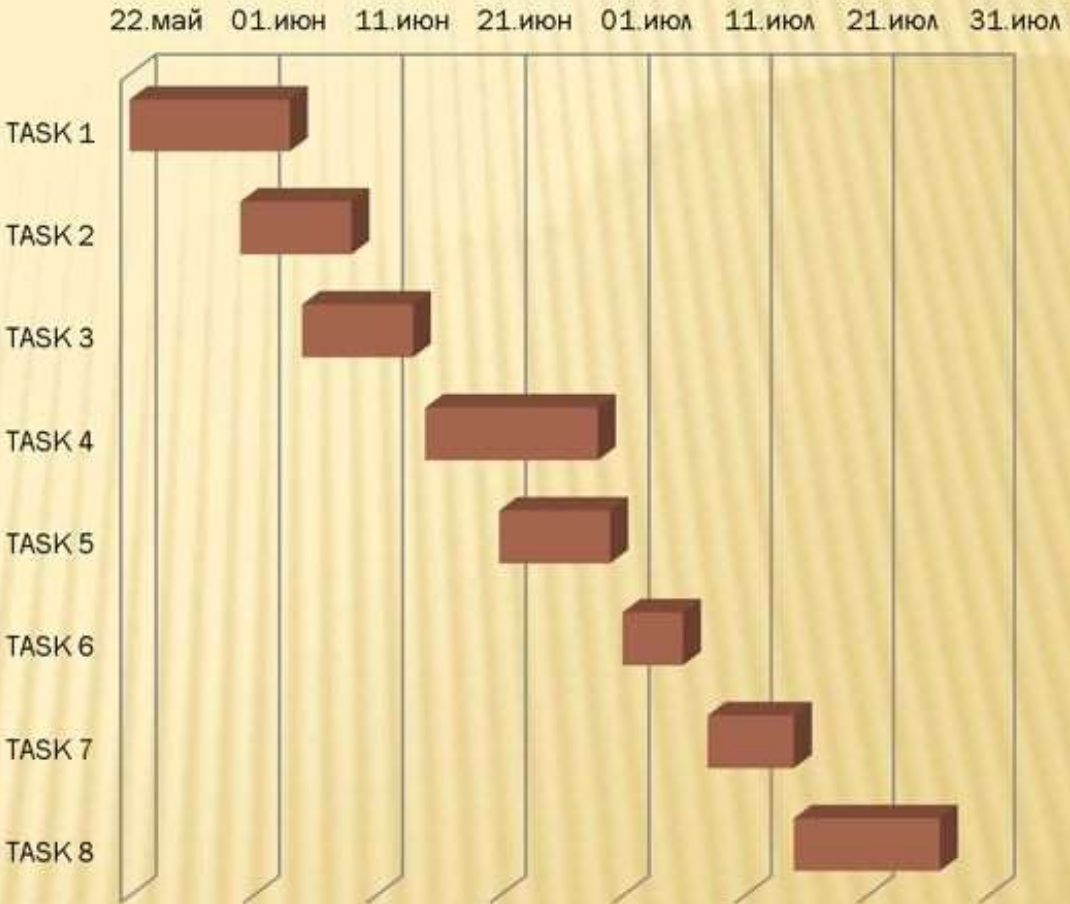
GANTT CHART

- ✖ It's a type of bar chart that illustrates a project schedule.
- ✖ It is the graphical representation of the duration of tasks against the progression of time.
- ✖ It shows the comparison between the planned and actual progress of job through several activities of departments.

AN EXAMPLE

TASK	START DAY	DAYS TO COMPLETE
TASK 1	22-May	13
TASK 2	31-May	9
TASK 3	5-Jun	9
TASK 4	15-Jun	14
TASK 5	21-Jun	9
TASK 6	1-Jul	5
TASK 7	8-Jul	7
TASK 8	15-Jul	12

GANTT CHART



	TASK 8	TASK 7	TASK 6	TASK 5	TASK 4	TASK 3	TASK 2	TASK 1
START DAY	15. июл	08. июл	01. июл	21. июн	15. июн	05. июн	31. май	22. май
DAYS TO COMPLETE	12	7	5	9	14	9	9	13